Supplementary information for:

Chapter 5 | Demographic history and temporal genomic erosion in Eurasian curlew *Numenius arquata*

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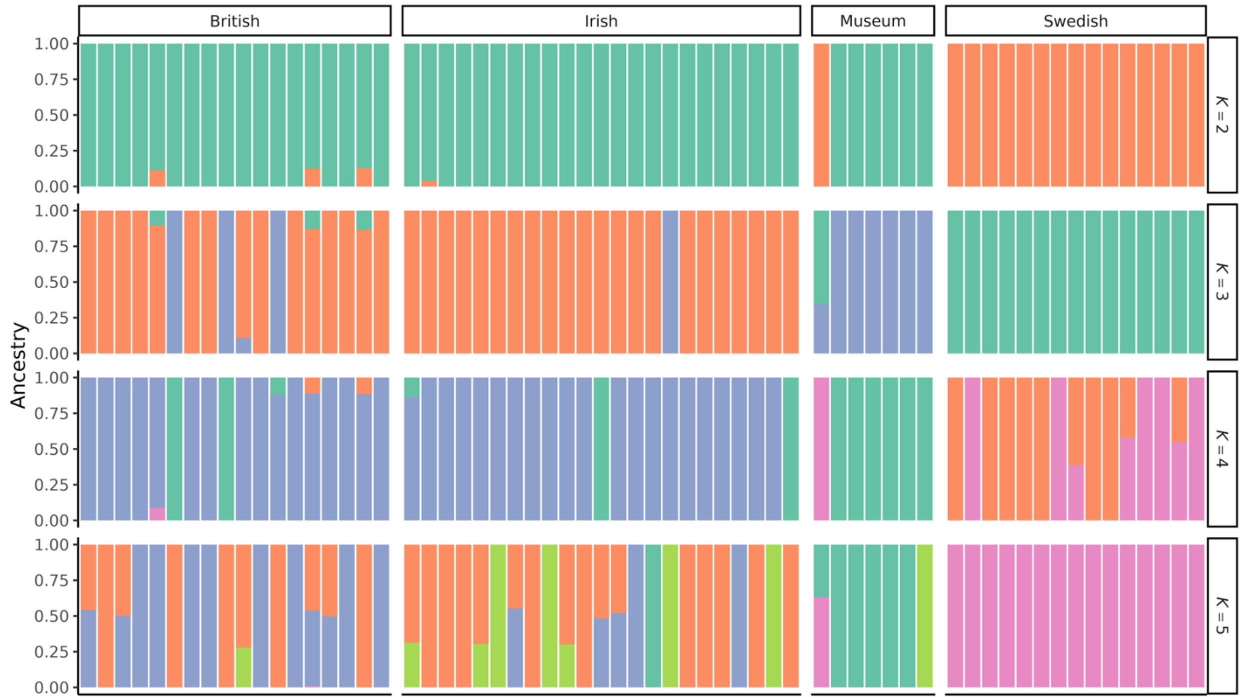

Figure S5.1: Bar plots of genetic ancestry generated with ADMIXTURE for K = 2–5 ancestral populations, for 63 Eurasian curlews (*Numenius arquata*), comprising contemporary samples from Ireland, Britain, and northern Sweden and seven historical museum specimens. Cross-validation error for each K is shown in Error! Reference source not found.**d**.

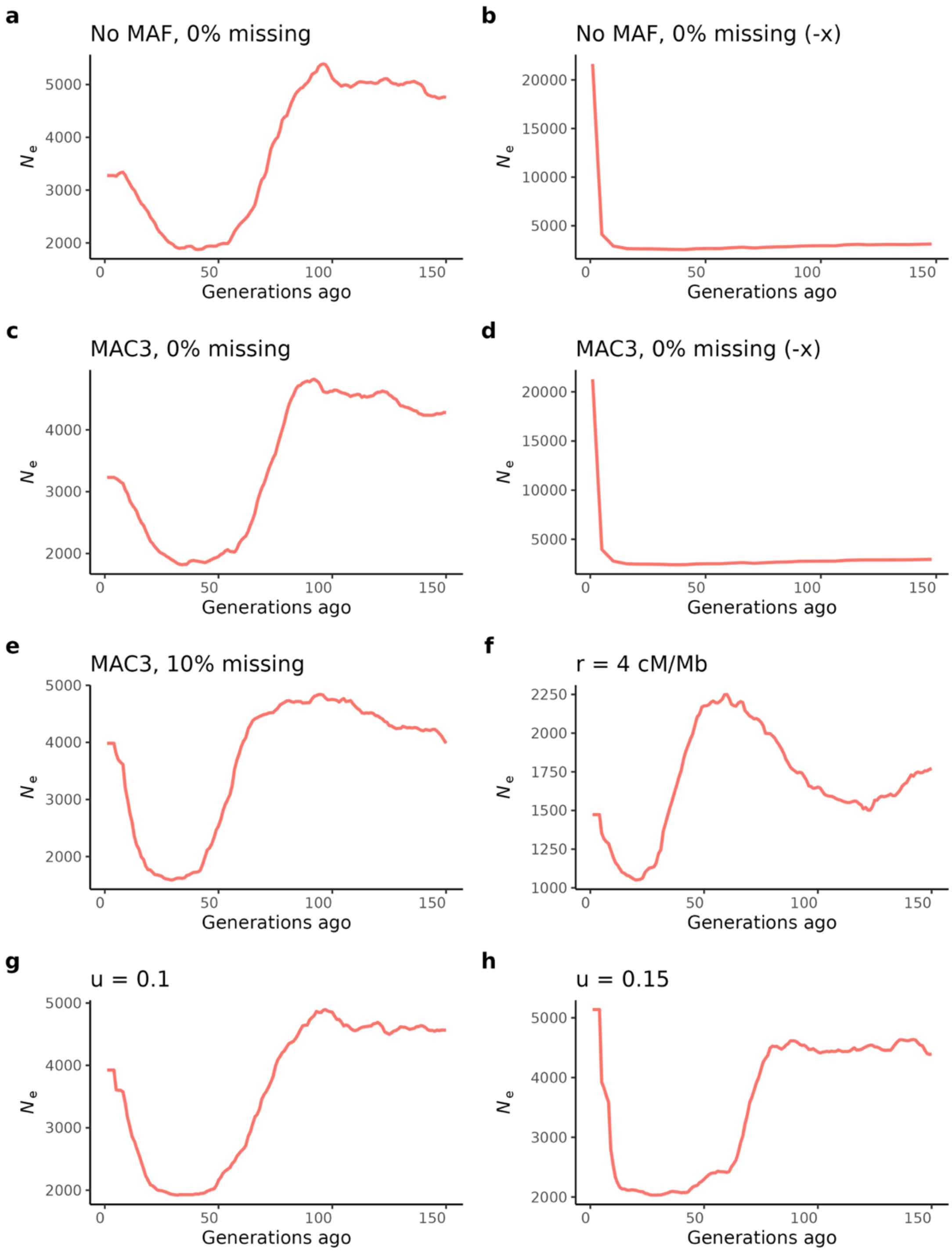

Figure S5.2: Supporting GONE2 runs for the Eurasian curlew (*Numenius arquata*) Ireland and Britain joined population under different parameter settings: (a) no minor allele filtering, 0% missing data allowed; (b) no minor allele filtering, 0% missing data allowed, structured population model (-x); (c) minor allele count (MAC) ≥ 3 filtering, 0% missing data allowed; (d) MAC3 filtering, 0% missing data allowed, structured population model (-x); (e) MAC3 filtering, 10% missing data allowed; (f) recombination rate of 4 cM/Mb; (g) upper bound of recombination rate u = 0.1; (h) u = 0.15. Default u is 0.05.

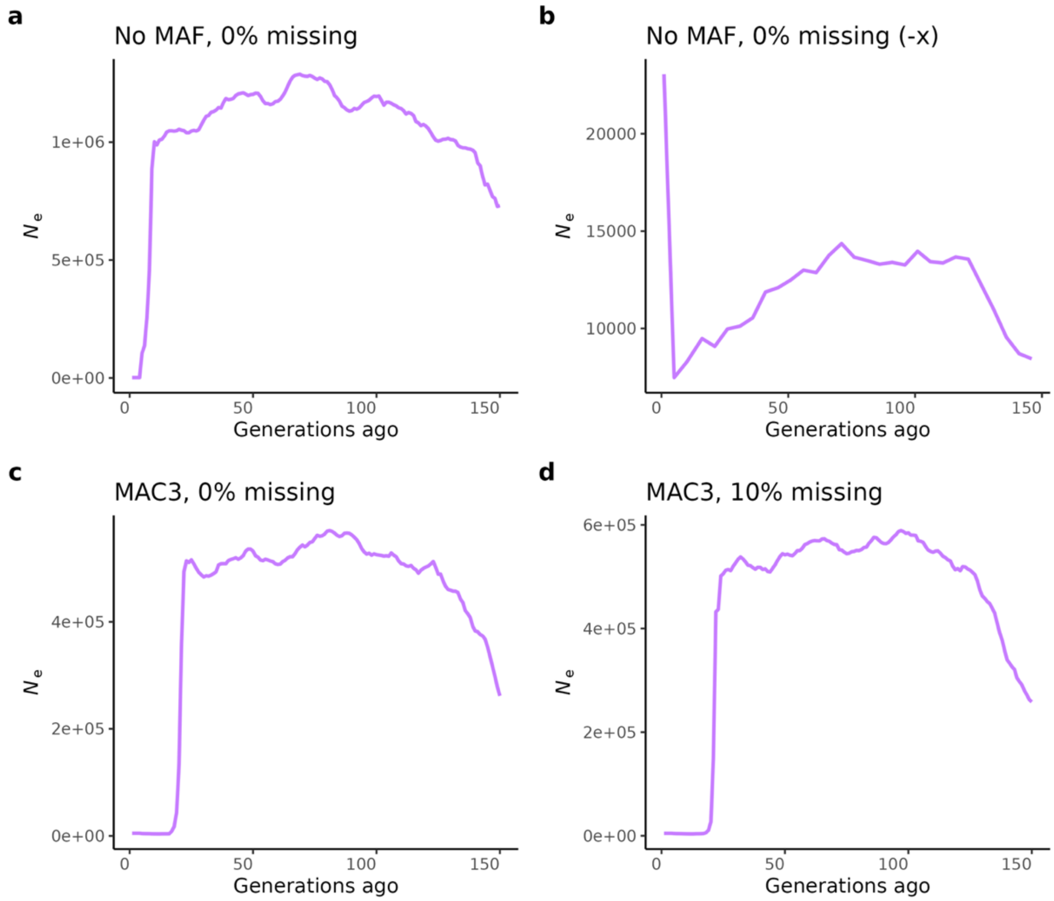

Figure S5.3: Supporting GONE2 runs for the Eurasian curlew (*Numenius arquata*) Swedish population under different parameter settings: (a) no minor allele filtering, 0% missing data allowed; (b) no minor allele filtering, 0% missing data allowed, structured population model (-x); (c) minor allele count (MAC) ≥ 3 filtering, 0% missing data allowed; (d) MAC3 filtering, 10% missing data allowed. A smaller set of runs is presented for the Swedish population compared to Ireland and Britain (**Figure S5.2**) as other runs did not converge.

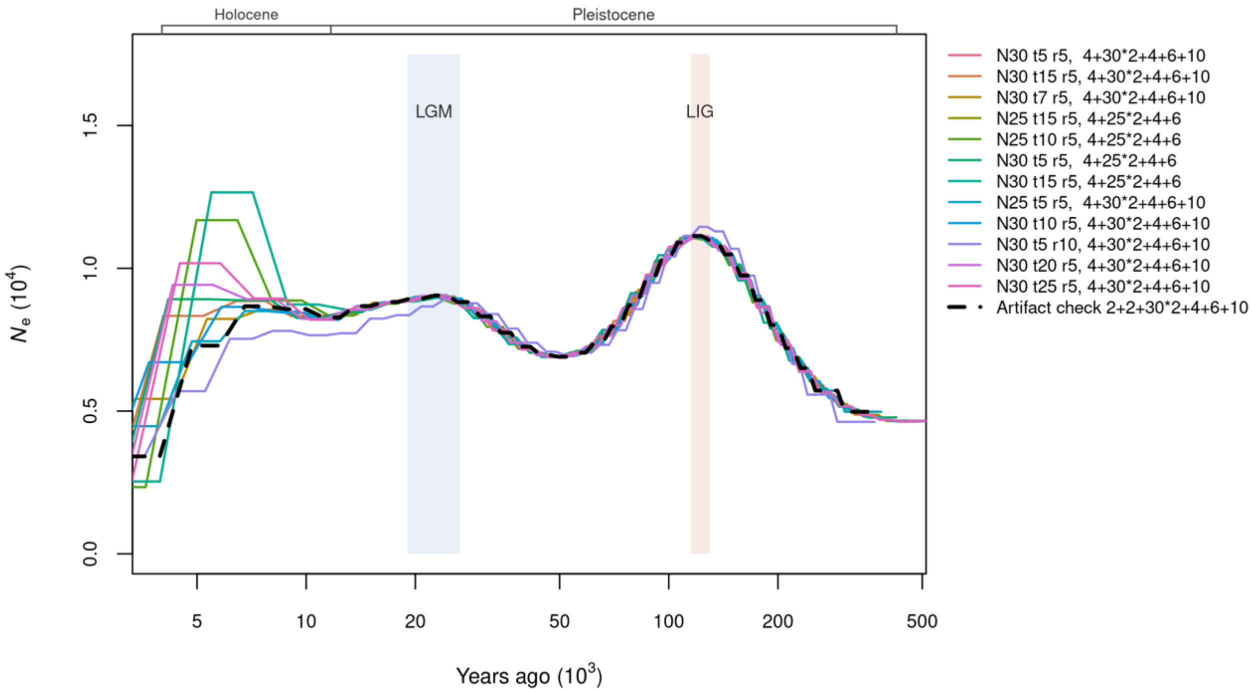

Figure S5.4: PSMC trajectories using 13 different sets of parameters to model effective population size (N_e) for a single high coverage British sample. The dashed black line indicates a false peak artefact check. Shaded bands indicate the Last Glacial Maximum (LGM) and Last Interglacial (LIG), and epoch brackets indicate the Pleistocene and Holocene.

Table S5.1: Normalised R_{XY} values for pairwise comparisons of contemporary and historical Eurasian curlew (*Numenius arquata*) populations. R_{XY} normalised to non-coding (modifier) sites, standard errors (SE) estimated using a weighted block jackknife over 115 physical blocks, and significance assessed with a single-sample t-test against the neutral expectation of 1. Significant results ($P < 0.05$) are shown in bold.

Comparison	**Impact**	R_{XY}	**SE**	P
Ireland–Britain	Moderate	1.002	0.010	0.825
Ireland–Britain	High	1.009	0.092	0.924
Ireland–Sweden	**Moderate**	**0.930**	**0.012**	**< 0.0001**
Ireland–Sweden	High	1.052	0.136	0.705
Britain – Sweden	**Moderate**	**0.928**	**0.017**	**< 0.0001**
Britain – Sweden	High	1.043	0.154	0.783
Ireland–Museum	**Moderate**	**0.951**	**0.013**	**0.0002**
Ireland–Museum	**High**	**0.676**	**0.063**	**< 0.0001**
Britain–Museum	**Moderate**	**0.949**	**0.017**	**0.003**
Britain–Museum	**High**	**0.668**	**0.083**	**0.0001**
Sweden–Museum	Moderate	1.023	0.015	0.131
Sweden–Museum	**High**	**0.647**	**0.075**	**< 0.0001**

Table S5.2: Genome-wide diversity estimates with 95% confidence intervals from block jackknife (1 Mb blocks) for four Eurasian curlew (*Numenius arquata*) cohorts. Ne estimated as θW/4μ (μ = 8.11 × 10^{-8})

Metric	Population	Mean	Standard Error	Lower CI	Upper CI
π	Ireland	2.91E-03	3.83E-05	2.83E-03	2.98E-03
	Britain	2.92E-03	3.86E-05	2.85E-03	3.00E-03
	Museum	3.19E-03	3.91E-05	3.11E-03	3.27E-03
	Sweden	3.09E-03	4.04E-05	3.01E-03	3.17E-03
Tajima's *D*	Ireland	0.179	0.006	0.167	0.192
	Britain	0.124	0.006	0.112	0.137
	Museum	-0.285	0.006	-0.297	-0.273
	Sweden	-0.210	0.006	-0.222	-0.199
θ_{W}	Ireland	2.70E-03	3.44E-05	2.63E-03	2.76E-03
	Britain	2.76E-03	3.52E-05	2.69E-03	2.83E-03
	Museum	3.29E-03	3.87E-05	3.22E-03	3.37E-03
	Sweden	3.22E-03	4.04E-05	3.14E-03	3.30E-03
N_e	Britain	8502	109	8289	8715
	Ireland	8314	106	8106	8521
	Museum	10147	119	9913	10381
	Sweden	9922	124	9678	10166

Table S5.3: Pairwise Wilcoxon rank-sum test p-values for genome-wide nucleotide diversity (π) and Watterson's theta (θ_W) among four Eurasian curlew (*Numenius arquata*) cohorts.

Comparison	Metric	*P*
Ireland–Britain	π	0.736
Ireland–Museum		**< 0.0001**
Ireland–Sweden		**0.001**
Britain–Museum		**< 0.0001**
Britain–Sweden		**0.004**
Museum–Sweden		0.069
Ireland–Britain	θ_W	0.214
Ireland–Museum		**< 0.0001**
Ireland–Sweden		**< 0.0001**
Britain–Museum		**< 0.0001**
Britain–Sweden		**< 0.0001**
Museum–Sweden		0.193